

Magnetic Field Effects on Photochemical Reaction in Mesoporous Silicates of MCM-41 under High Magnetic Fields up to 5 T

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Magnetic field effects (MFEs) on photoinduced hydrogen abstraction reaction of xanthone with xanthene were investigated in 2-propanol confined in mesoporous silicates of MCM-41 under high magnetic fields up to 5 T. The observed MFEs were explained by the spin relaxation mechanism associated with the cage effects on diffusion of the radical pairs in nanopores of MCM-41. The qualitative analysis using the stochastic Liouville equation revealed that the motions of the radical pairs are not restricted in the one dimension, although the MCM-41 used in the present study has the one-dimensional pore with the diameter of 1.7 or 2.7 nm. The effective viscosities for rotation of the solute molecule in the pores were estimated to be 5–10 cP.

Introduction

Mesoporous silicates have pore diameters from 2 to 50 nm.¹ Because of their large specific surface areas and tunable pore sizes, mesoporous silicates are expected to be new materials for catalysis.^{2–5} Also, various applications of mesoporous silicates to nanochemistry such as synthesis of nanocomposites,^{6–8} size-selective separation,^{9,10} and drug delivery^{11,12} have been reported. The one-dimensional pores in mesoporous silicates have received considerable attention not only for development of functional materials but also for molecular dynamics of a one-dimensional diffusion model. Molecular diffusions in mesoporous silicates have been studied with several techniques in liquid media, including absorption and fluorescence spectroscopy,^{13,14} neutron scattering,¹⁵ and electron paramagnetic resonance (EPR) spectroscopy.¹⁶ These studies provide valuable information on molecular diffusion in mesoporous silicates. Especially, it has been suggested that solutions in mesoporous silicates collectively flow inside of the pores whose diameters are larger than 2 nm.

Magnetic field effects (MFEs) on photochemical reactions through radical pairs (RPs) and biradicals have received considerable attention during the past 3 decades.^{17–19} Magnetic fields interact with the electron spins of RPs, and thus spin conversion in the RPs is influenced by the fields. The lifetime of the RPs and the yield of the escaped radicals show appreciable MFEs. We have recently demonstrated that the MFE studies on RPs provide valuable information on their kinetics and dynamics and in particular on aspects of the environments around RPs.²⁰ Because interactions of electron spins in RPs are limited to within a few nanometers, the MFEs are affected by the microenvironment in the vicinity of RPs rather than the macroscopic environments. Therefore, the microenvironments in nanopores of mesoporous silicates can be probed by the MFEs.

Okazaki et al. reported MFEs on RPs generated from photoinduced hydrogen abstract reaction between xanthone (XO) and xanthene (XH₂) in mesoporous silicates. The RPs were suggested to be confined in the pore for several tens of

microseconds.²¹ Clustering of solvent alcohols inside pores by hydrogen-bonding interactions was proposed as an origin of the observed confinement effects. However, their study was limited to magnetic fields lower than 500 mT and the detailed mechanism on the MFEs has not been discussed.

In this paper, we studied the MFEs on the same photochemical reaction of XO with XH₂ in 2-propanol confined in mesoporous silicates of MCM-41 using the superconducting magnet that can generate high magnetic fields up to 5 T. MCM-41 is a mesoporous silicate having regularly ordered hexagonal pore arrangement and narrow pore size distributions.^{22–24} For comparison, the MFEs on the same photochemical reaction were also measured in sodium dodecyl sulfate (SDS) micellar solutions by using nanosecond laser flash photolysis. In high magnetic fields, saturations of MFEs were observed in MCM-41, which enable us to analyze the experimental data qualitatively using the stochastic Liouville equation (SLE). Effective viscosities for radical diffusion in nanopores of MCM-41 were estimated to be 5–10 cP, which were larger than the macroviscosity of the solvent. We propose heterogeneous solvent clustering to account for the observed MFEs in MCM-41.

Experimental Section

Materials. Mesoporous silicates of MCM-41 were synthesized by using water glass (Na₂O, 9.5 wt %; SiO₂, 28.6 wt %; Nippon Chemical Industry Co.), sulfuric acid, pure water, and alkyltrimethyl ammoniumbromide [(CH₃)₃NC_nH_{2n+1}Br] as a template.²⁵ Lengths of the alkyl groups (*n*) are 10 and 16 for MCM (1.7) and MCM (2.7), respectively. Here, MCM (*d_p*) indicates MCM-41 with cylindrical pores with diameter *d_p* in nanometers. The MCM-41 as synthesized was calcined for 6 h at 813 K. The physical properties of the MCM-41 materials are listed in Table S1 (Supporting Information).

Xanthen-9-one [xanthone (XO), Cica] and xanthene (XH₂, Cica) were recrystallized from ethanol. SDS (Cica) and 2-propanol (2-PrOH, Cica) were used as received. Water used was superpure water (Millipore, Simpli Lab).

Steady-State Photolysis. XO and XH₂ were dissolved in 2-PrOH with the concentrations of 1.0×10^{-3} and 3.0×10^{-3} mol dm⁻³, respectively. Sample solutions were nitrogen-bubbled for 1.5 h prior to measurements. The sample solution was

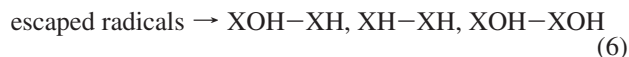
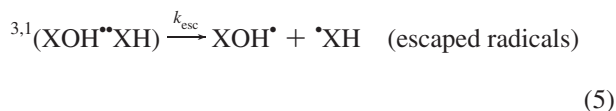
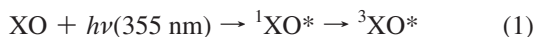
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pumped (Shimadzu, LC-9A) through a glass column (EYELA, i.d. = 3 mm, length = 150 mm) packed with MCM-41. The pressure at the inlet of the column was about 2×10^6 N/m² with a flow rate of 0.2 mL/min. The glass column was set at the center of space in a superconductive magnet (Toshiba, TM-5HSP). The solution in the column was irradiated with a xenon arc lamp (Ushio, UXL500D) through a water filter and a cut-off filter (HOYA, UV-32). After the irradiation, the sample solutions collected for the high-performance liquid chromatography (HPLC) analysis were cooled immediately with iced water to prevent the decomposition of the product molecules. The HPLC unit (Jasco, X-LC) was equipped with a UV/vis detector (Jasco, 3075UV) and a reversed phase column (Jasco, X-PressPak C18S). The eluent solvents were acetonitrile (AN) and water (gradient: from 50 to 95% AN). All measurements were performed at 297 K. Quantitative analysis of products was carried out by HPLC with biphenyl as an internal standard.

Nanosecond Laser Flash Photolysis. Nanosecond laser flash photolysis was carried out using an apparatus described elsewhere.²⁰ The third harmonic (355 nm) of nanosecond Nd:YAG laser (Quanta-Ray, GCR-130-10 or INDI) was used as an excitation light source. XO, XH₂, and SDS dissolved in pure water with the concentrations of 5×10^{-4} , 1.0×10^{-3} , and 8×10^{-2} mol dm⁻³, respectively. Argon-bubbled sample solution was circulated to the quartz cell placed at the center of magnets. The magnetic fields up to 1.5 T were provided by an electromagnet (Tokin, SEE-10W) and up to 4 T by a homemade pulsed magnet.²⁶

Results and Discussion

Reaction Scheme. The following reactions occur by the photoexcitation of XO in the presence of XH₂ in MCM-41 filled with alcohols²¹ and in an SDS micellar solution,²⁷



Here, ${}^1\text{XO}^*$ and ${}^3\text{XO}^*$ represent the singlet and triplet excited states of XO, respectively. XOH[•] and [•]XH represent the xanthone ketyl and xanthenyl radicals, respectively. ${}^1(\text{XOH}^{\bullet}\text{XH})$ and ${}^3(\text{XOH}^{\bullet}\text{XH})$ denote singlet and triplet RPs composed of XOH[•] and [•]XH, respectively. Upon irradiation of XO, ${}^3\text{XO}^*$ is immediately generated via the fast intersystem crossing of ${}^1\text{XO}^*$ (eq 1). The triplet RP is given by a hydrogen abstraction reaction of ${}^3\text{XO}^*$ from XH₂ (eq 2). In the presence of 2-PrOH, a hydrogen abstraction reaction of ${}^3\text{XO}^*$ from 2-PrOH to form

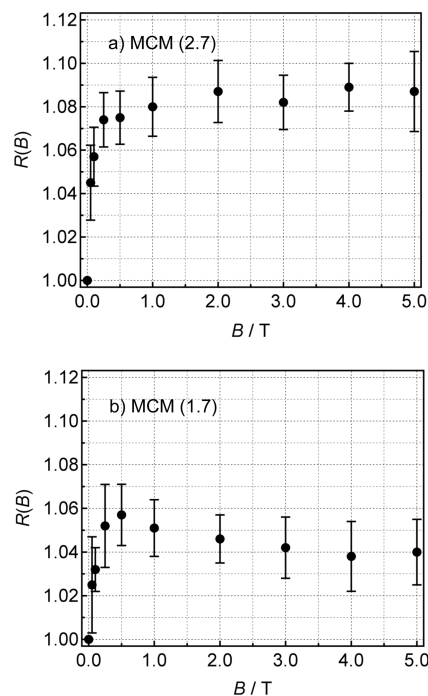


Figure 1. Magnetic field effects on the yield of bixanthyI generated by the photoinduced hydrogen abstraction reaction of xanthone (1.0×10^{-3} mol dm⁻³) with xanthene (3.0×10^{-3} mol dm⁻³) in 2-propanol confined in (a) MCM (2.7) and (b) MCM (1.7). Experimental errors plotted in the figure are standard deviations calculated from five or seven HPLC measurements of a sample at a field.

2-propanol radical (2-PrOH[•]) is also possible. 2-PrOH[•] is however unstable and finally gives [•]XH after the radical migration when the concentration of XH₂ is sufficiently high. The spin conversions between ${}^1(\text{XOH}^{\bullet}\text{XH})$ and ${}^3(\text{XOH}^{\bullet}\text{XH})$ depend on the magnetic field (*B*). The recombination reaction occurs from the singlet RPs with a rate constant k_{rec} (eq 4), giving the cage product (XOH-XH) while the radicals escape from the singlet and triplet RPs with a rate constant k_{esc} (eq 5). The escaped radicals react with the other and give the escape products (XOH-XH, XH-XH, XOH-XOH).

MFES Observed in MCM-41. The yield (*Y*) of bixanthyI (XH-XH), which is one of the escaped products for the geminate RPs, was quantitatively analyzed by HPLC. The relative magnetic field effect ($R(B)$) on the yield of XH-XH was calculated as $R(B) = Y(B)/Y(0 \text{ T})$. The observed $R(B)$ values with MCM (2.7) and with MCM (1.7) are plotted against *B* in Figure 1, respectively. For both MCM-41, the $R(B)$ values increased with increasing *B* in the range of $0 < B \leq 0.2$ – 0.5 T. These MFES qualitatively agree with the MFES reported by Okazaki et al. and is explained by the spin relaxation mechanism.²¹

Above 0.5 T, the $R(B)$ values observed with MCM (2.7) and MCM (1.7) showed the different *B*-dependence. For MCM (2.7), the $R(B)$ values were once saturated in the range of $0.25 < B \leq 0.5$ T and gradually increased again with increasing *B* in $0.5 < B \leq 2$ T. The $R(B)$ values were then saturated in the range of $2 < B \leq 5$ T. For MCM (1.7), however, the $R(B)$ values gradually decreased with increasing *B* in the range of $0.5 < B \leq 3$ T and were saturated in $3 < B \leq 5$ T. Although the experimental errors calculated from five or seven HPLC measurements of a sample at a field are somewhat large, the difference of *B*-dependence observed with MCM (2.7) and MCM (1.7) were constantly shown in five different series of measurements ($0 \leq B \leq 5$ T).

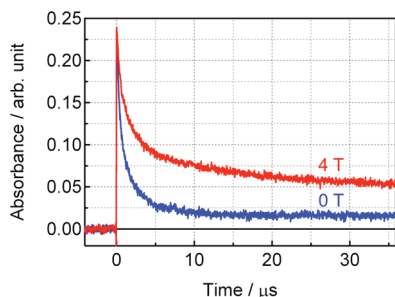


Figure 2. $A(t)$ curves observed at 490 nm in an SDS micellar solution containing xanthone (0.5×10^{-3} mol dm $^{-3}$) and xanthene (1.0×10^{-3} mol dm $^{-3}$) in the absence (blue line) and presence (red line) of a magnetic field of 4 T.

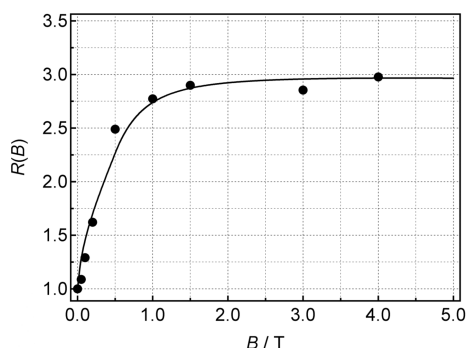


Figure 3. Magnetic field effects on the yield of escaped xanthone ketyl radicals observed at 490 nm in SDS micellar solution. Solid line shows a simulated curve obtained from the SLE analysis.

MFES Observed in SDS Micellar Solution. To investigate the MFES on the present reaction in an SDS micellar solution, the yield of the escaped radical XOH^* , which has the transient absorption band around 500 nm,^{27,28} was monitored with nanosecond laser flash photolysis under various magnetic fields. The magnetic field dependence on the time profiles of the transient absorption ($A(t)$) were measured at 490 nm in the absence and presence of magnetic fields. Typically, the $A(t)$ curves observed at 0 and 4 T are shown in Figure 2. The $A(t)$ curves had three decay components. From the analysis of the $A(t)$ curve by a three exponential fit, the lifetimes of fast, middle, and slow decay components obtained were 0.33, 1.6, and 14 μs , respectively. The fast (lifetime (τ) = 0.33 μs) and middle (τ = 1.6 μs) decay components can be ascribed to the decay of $^3\text{XO}^*$ and decay of $^{3,1}(\text{XOH}^* \cdot \text{XH})$, respectively. The slow one (τ = 14 μs) is ascribed to the decay of escaped XOH^* . Apparently, the yield of the escaped XOH^* is affected by the magnetic field. The $R(B)$ values on the yield of XOH^* was obtained by using the transient absorbance at the delay time of 15 μs after laser excitation: $R(B) = Y(B)/Y(0 \text{ T}) = A(15 \mu\text{s}, B \text{ T})/A(15 \mu\text{s}, 0 \text{ T})$. Figure 3 shows a magnetic field dependence of $R(B)$ observed in the SDS micellar solution. The $R(B)$ values increased with increasing B in the region of $0 < B \leq 1.5$ T and were saturated in $1.5 < B \leq 4$ T. In the SDS micellar solution, the observed MFE is also explained by the spin-relaxation mechanism.²⁷ The experimental errors in the nanosecond laser flash photolysis (Figure 3) were estimated to be within 2–3%.

Analysis of the Stochastic Liouville Equation. Both MFES observed in MCM-41 and the SDS micellar solution can be explained by the spin relaxation mechanism. To investigate the influence of the radical diffusion in one-dimensional pore on the MFES, we performed qualitative analysis of the observed MFES by using the SLE.^{29,30} The details of the SLE analysis are described in the Supporting Information and in our previous

paper.^{20c} The SLE analysis includes the effects of spin–spin interactions, molecular diffusion, recombination reactions, and spin relaxations. The spin Hamiltonian consists of the exchange interaction (J), the hyperfine interaction, and the Zeeman interactions caused by the magnetic field. The radical diffusion, which is assumed to be simple Brownian motion, is treated with the finite difference technique and the mutual diffusion constant (D). D is represented as a sum of the diffusion coefficients for radical a and b ($D = D_a + D_b$).

In the three-dimensional diffusion, the diffusion coefficient for each radical is represented with the Stokes–Einstein equation as follows,

$$D_a(\eta) = \frac{k_B T}{6\pi\eta d_a}, \quad D_b(\eta) = \frac{k_B T}{6\pi\eta d_b} \quad (7)$$

where d_a and d_b are radii of each radical and η is the solvent viscosity, respectively.

The cage effects on the RP diffusion have been reported in MCM-41²¹ and in the SDS micellar solution.²⁷ We, therefore, employed the cage model to analyze the experimental data using the one-dimensional and three-dimensional diffusion model. In the cage model, one radical is fixed to the center of the cage, while the other radical diffuses freely in the cage. The RPs recombine at the contact or escape from the cage with a certain probability (P_{esc}) from the cage at the interface.

The main feature of the observed MFES is explained by the spin relaxation mechanism. For the magnetic field dependent spin relaxations, we took account of the spin relaxations caused by the dipole–dipole spin relaxation, the anisotropies (δg) of the g -factors, and the anisotropy (δA) of the hyperfine interaction. For the dipole–dipole spin relaxation, the following longitudinal relaxation rate constant ($1/T_1$) due to the dipole–dipole interactions is used,¹⁹

$$\frac{1}{T_1} = \frac{\mu_B^4 g_a^2 g_b^2}{10\hbar^2 r^6} \times \frac{3\tau_{ab}}{1 + (\omega + \omega_{\text{DDI}})^2 (\tau_{ab})^2} \quad (8a)$$

$$\omega_{\text{DDI}} = \frac{3}{2\sqrt{5}} \times \frac{\mu_B^2 g_a g_b}{\hbar r^3} \quad (8b)$$

where g_a and g_b are the g -factors for radicals a and b , respectively. r is the center-to-center separation between the radicals. ω is the Larmor frequency for the unpaired electron spin and is dependent on B ; $\omega = g\mu_B\hbar^{-1}B$. ω_{DDI} corresponds to the averaged energy splitting due to the dipole–dipole interactions for the spin-sublevel states in the triplet state.³¹ τ_{ab} is the rotational correlation time of the vector directing radical a to radical b . Equation 8a cannot account for the spin relaxation rate at the magnetic field lower than 0.01 T.³² Previously, we estimated the spin relaxation rate at the zero magnetic field with the comparison of the data obtained at the magnetic fields of up to 30 T.²⁶ The estimated spin relaxation rate at the zero magnetic field was on the order of 10^7 s^{-1} . At the magnetic field lower than the 0.01 T, we used the rate of $1 \times 10^7 \text{ s}^{-1}$ for the spin relaxation due to the dipole–dipole interactions in the overall cage.

The $1/T_1$ values for the longitudinal relaxation caused by δg and δA are represented as¹⁹

$$\frac{1}{T_1} = \frac{3(g'_a g'_a) \mu_B^2 B^2 + (A':A')}{30\hbar^2} \times \frac{\tau_a}{1 + \omega^2 \tau_a^2} + \frac{3(g'_b g'_b) \mu_B^2 B^2}{30\hbar^2} \times \frac{\tau_b}{1 + \omega^2 \tau_b^2} \quad (9)$$

where $(g':g')$ and $(A':A')$ are the anisotropies for the g -factor and the hyperfine coupling constant, respectively. τ_a and τ_b are the rotational correlation times for radicals a and b , respectively. In the presence of the large J value in the closed RP, the transverse relaxation rate constant ($1/T_2$) for S–T₀ relaxation is represented as follows: For $|2J(r)| > \{[3(g'_a g'_a) \mu_B^2 B^2 + (A':A')]/30\hbar^2\}^{1/2}$,

$$\frac{1}{T_2} = \frac{3(g'_a g'_a) \mu_B^2 B^2 + (A':A')}{30\hbar^2} \times \frac{\tau_a}{1 + [2J(r)]^2 \tau_a^2} + \frac{3(g'_b g'_b) \mu_B^2 B^2}{30\hbar^2} \times \frac{\tau_b}{1 + [2J(r)]^2 \tau_b^2} \quad (10)$$

When the S and T₀ state is nearly degenerated, however, $1/T_2$ is represented as follows: For $|2J(r)| < \{[3(g'_a g'_a) \mu_B^2 B^2 + (A':A')]/30\hbar^2\}^{1/2}$,

$$\frac{1}{T_2} = \frac{3(g'_a g'_a) \mu_B^2 B^2 + (A':A')}{30\hbar^2} \times 4\tau_a + \frac{3(g'_b g'_b) \mu_B^2 B^2}{30\hbar^2} \times 4\tau_b \quad (11)$$

In the homogeneous solvents, the rotational correlation time (τ_{ab}) for RP and for radicals a and b (τ_a , τ_b) are correlated with η ,

$$\tau_{ab} = \frac{4\pi\eta r^3}{3k_B T} \quad (12a)$$

$$\tau_a = \frac{4\pi\eta d_a^3}{3k_B T} \quad (12b)$$

$$\tau_b = \frac{4\pi\eta d_b^3}{3k_B T} \quad (12c)$$

For both solutes, we used the value of $d_a = d_b = 0.4$ nm in the SLE analysis.

Simulations for SDS Micellar Solution. We first performed the SLE analysis on the MFEs observed with the SDS micelle where the cage parameters are well-known. The core and hydrodynamic diameters for the SDS micelle were estimated to be 3.3 and 4.1 nm from its alkyl chain length, respectively.^{33,34} The η values inside of SDS have been reported to be 9–19 cP.^{35–38} The results of the SLE analysis on the MFEs observed in the SDS micellar solution were depicted in Figure 3 by a solid line. The parameters used in the SLE analysis are listed in Table 1. The MFEs observed experimentally in the SDS micellar solutions were well reproduced by the SLE analysis and is mainly explained by the magnetic field dependent spin relaxation caused by the dipole–dipole interactions. The η value

of 25 cP used in the present SLE analysis was somewhat larger than reported values of 9–19 cP. It has been known that the experimentally observed η values depend on the employed probe molecules.^{33,39} The relatively large η value obtained in the present study may reflect the relatively strong interaction between the intermediate radicals and the SDS micelle.

Simulations for MCM-41 with One-Dimensional Diffusion Model. We first performed the SLE simulations for MCM-41 by using the one-dimensional diffusion model because MCM-41 has regularly ordered hexagonal one-dimensional pores. The SLE analysis on the MFEs observed in MCM-41 was performed with the k_{rec} value of 3×10^8 s^{−1} which was the same with the SDS micelle. In the SLE analysis, we still employed eqs 7 and 12a–c for convenience, although physical meaning of η and D in the one-dimensional diffusion model is different from those in

the three-dimensional model. In our experimental conditions, the photochemical reactions occurs both inside and outside MCM-41. The absolute values of MFEs therefore are somewhat arbitrary. In the simulation, the calculated results were corrected by using the correlation factor (γ):

$$R(B) = \gamma \times \{Y(B)/Y(0 \text{ T}) - 1\} + 1 \quad (13)$$

Typical γ value used in the SLE analysis was in the range of 0.1–0.2.

Recent theoretical study on chemically induced dynamic electron polarization (CIDEP) reported by Lukuzen et al. suggested that the dipole–dipole interaction in RPs is not fluctuated when the RPs motion is restricted in one-dimensional space.⁴⁰ In such condition, the MFE due to the dipole–dipole spin relaxation mechanism is not expected. To clarify the effect of the dipole–dipole interactions on the present MFEs, the SLE calculations were performed with and without the spin relaxation due to the dipole–dipole interactions. Figure 4 shows calculated $R(B)$ for MCM (2.7) and MCM (1.7) using the one-dimensional diffusion model together with the experimental results (filled circle). The parameters used in the SLE analysis are listed in Table 1. Red lines show calculations including the spin relaxation caused by the dipole–dipole interaction whereas the blue lines are calculations without the spin relaxation due to the dipole–dipole interactions. As can be seen in Figure 4, it is noteworthy that the spin relaxation due to the dipole–dipole interaction is necessary to reproduce the experimental results. This fact clearly shows that the RP diffusion in pores of MCM-41 is not the one-dimensional diffusion. When the spin relaxation due to the dipole–dipole interaction is not included in the calculation (blue lines), the calculated $R(B)$ shows a steep increase in range of $0 < B < 0.2$ T with increasing B , gradual decrease in $0.2 < B < 4$ T, and the saturation in $4 < B < 5$ T. These MFEs are not in agreement with the experimental results. The steep increase in the low magnetic field is explained by the longitudinal spin relaxation due to the fluctuation of δA while the gradual decrease is interpreted by the transverse spin relaxation caused by δg . The relatively large dipole–dipole interaction induces the spin relaxation in the higher magnetic field, giving the gentle increase of $R(B)$ with increasing B as was observed in the experiments ($0.5 < B \leq 2$ T for MCM(2.7) and $0 < B \leq 0.5$ T for MCM(1.7)). The calculated results indicate that the dipole–dipole interaction is fluctuated in MCM-41 and the RP diffusion is not restricted in one dimension.

Next, the observed magnetic field dependence is discussed in detail. As for MCM (2.7), the saturation of the MFEs in the range of $0.25 < B \leq 0.5$ T is interpreted by the spin relaxation

TABLE 1: Parameters Used for the SLE Analysis; Recombination Reaction Rate (k_{rec}) at the Closest Radical–Radical Distance, Escape Probability (P_{esc}) at the Interface, Rotational Correlation Time (τ), Viscosity (η) in Cages, and Length (L) and Diameter (d) for the Cage in One-Dimensional and Three-Dimensional Diffusion Model

system	$k_{\text{rec}}/\text{s}^{-1}$	1D Diffusion Model				3D Diffusion Model			
		P_{esc}	$\tau_a, \tau_b/\text{s}$	η/cP^a	L/nm	P_{esc}	$\tau_a, \tau_b/\text{s}$	η/cP	d/nm
MCM (1.7)	3×10^8	4.6×10^{-3}	3.9×10^{-10}	6	5.8	2.3×10^{-4}	4.6×10^{-10}	7	3.2
MCM (2.7)	3×10^8	3.5×10^{-3}	5.2×10^{-10}	8	7.8	1.5×10^{-4}	6.5×10^{-10}	10	3.4
SDS	3×10^8					1.2×10^{-4}	1.6×10^{-9}	25	3.6

^a Estimated from the rotational correlation times for radicals and the values of $d_a = d_b = 0.4$ nm with eq 12b and 12c.

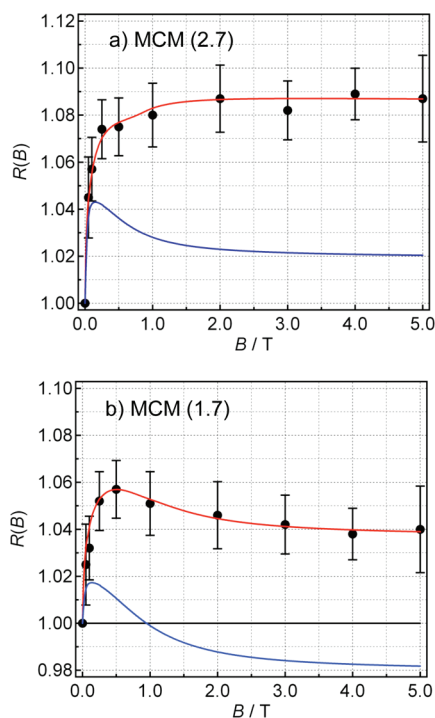


Figure 4. Magnetic field effect on the yield of escaped radicals calculated from the SLE analysis with the one-dimensional diffusion model for (a) MCM (2.7) and (b) MCM (1.7). Red lines show simulations with the dipole–dipole spin relaxation while blue lines indicate simulations without dipole–dipole spin relaxation. Filled circles are corresponding experimental data.

due to δg . The gradual increase of MFEs in the range of $0.5 < B \leq 2$ T is explained by the spin relaxation caused by the dipole–dipole interaction. For MCM (1.7), the gradual increase of MFEs in the range of $0 < B \leq 0.5$ T is explained by the spin relaxation caused by the dipole–dipole interaction while the gradual decrease of MFEs in the range of $0.5 < B \leq 3$ T is explained by the spin relaxation caused by δg . These spin relaxation processes are associated with the molecular rotations as can be seen in eqs 9–13. Figure 5 shows $R(B)$ calculated by the one-dimensional diffusion model with various η values for MCM (2.7) and for MCM (1.7), respectively. The shapes of the MFEs are sensitive to η , indicating that the effective viscosity for the radical rotation can be extracted from the observation of the MFEs. From the SLE analysis, the η value effective for the solute molecule rotation is estimated to be 6–7 cP for MCM (1.7) and 8–10 cP for MCM (2.7), respectively.

Simulations for MCM-41 with Three-Dimensional Diffusion Model. From the SLE analysis using the one-dimensional diffusion model, we found that the spin relaxation due to the dipole–dipole interaction is necessary to reproduce the experimental results and the RP rotation is not restricted in the one dimension in MCM-41. Therefore, the one-dimensional diffusion model is not completely adequate to explain the MFEs observed

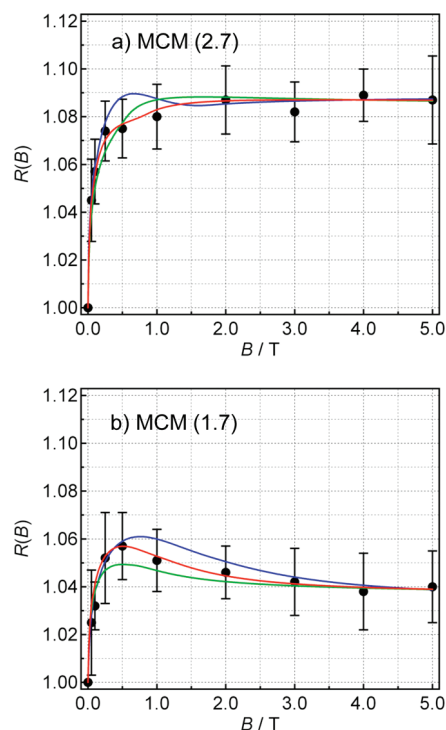


Figure 5. (a) Magnetic field effect on the yield of escaped radicals calculated with $\eta = 2$ cP (green line), $\eta = 8$ cP (red line), and $\eta = 20$ cP (blue line) with the one-dimensional diffusion model for MCM (2.7). (b) Magnetic field effect on the yield of escaped radicals calculated with $\eta = 2$ cP (green line), $\eta = 6$ cP (red line), and $\eta = 15$ cP (blue line) with the one-dimensional diffusion model for MCM (1.7).

with MCM-41. Then we performed a similar SLE analysis by using the three-dimensional cage model. The experimental results can also be reproduced by the three-dimensional model. The calculated $R(B)$ for MCM (2.7) and MCM (1.7) together with the experimental results are shown in Figure S1 (Supporting Information). The parameters used in the SLE analysis with the three-dimensional diffusion model are also listed in Table 1. In the three-dimensional diffusion model, the cage parameters of P_{esc} and d for MCM-41 are similar to that obtained in the SDS micellar solution. The η value for MCM-41 is, however, different from that for the SDS micellar solution. The larger η value gives the longer lifetime of RPs. Thus, the present SLE analysis indicates that the lifetime of RP in MCM-41 is shorter than that in the SDS micellar solution.

Solute Molecule Diffusion in MCM-41. The parameters of P_{esc} and the cage size (L and d) are varied with the models used in the SLE simulation. The quite similar η value is however obtained in both diffusion models. The η value determines the rotational correlation time of radicals and RPs and affects the magnetic field dependence of the spin relaxations in the SLE. The shapes of the calculated MFEs are therefore similarly influenced by the η value in both models. As listed in Table 1, the obtained η values in MCM-41 are larger than the viscosity



Figure 6. Schematic representation of a model for the radical pair diffusion in nanoporous silicates. Region A: solvent molecules are strongly aggregated by hydrogen bonding, prohibiting the penetration of solute molecules. Region B: solute molecules (filled circles) freely diffuse in the three-dimensional space.

(2 cP) for the bulk solutions of 2-PrOH. The results are consistent with the solvent dynamics study reported by Kamijo et al.^{14b} They reported that the solvent relaxation time in MCM-41 becomes much larger than those observed in bulk solutions when ethanol and butyl alcohol are used for the solutions. The slow solvent relaxation time observed was interpreted with the relatively rigid assembly of alcohols by hydrogen bonding in MCM-41. In the present study, 2-PrOH is expected to form hydrogen bonding with each other and diffuse collectively in MCM-41. Such effects cause the slow solute molecular rotation in MCM-41 and give viscosity higher than the bulk solution.

For the translational diffusion of solute molecules, the SLE analysis revealed that the RP diffusion in the nanopore is not restricted in the one dimension. Fluorescence study suggested that the structure of MCM-41 has defects, which allow the solute molecule to diffuse to other pores, resulting in the three-dimensional diffusion of RPs.¹³ The NMR studies indicate that the shape of the nanopore of MCM-41 is not straight, which also causes the three-dimensional diffusion of solute molecules.^{41,42} In these situations, however, the re-encounter probability of RP is not high and one cannot expect the cage effects on the RP diffusions. In the present study, we observed the clear MFEs and the cage effects on the RP diffusion. The effects of structural defects and distortion of MCM-41 for the RP diffusion are therefore excluded as an origin of the three-dimensional diffusion of RP observed in the present study.

The observation of the cage effects in MCM-41 are consistent with the MFE study in the low magnetic field region reported by Okazaki et al.²¹ In their model, the geminate RPs are separated by the solvent cluster and do not recombine in nanopores.²¹ The recombination reactions occur after the RPs flow to the outside of the nanopore. Here we propose a somewhat different model to account for the MFEs observed in MCM-41, though the origin of the cage effects in MCM-41 is unclear at present. In our model, we consider the heterogeneous solvent clustering due to the solute molecules in MCM-41. The model is depicted in Figure 6. Black parts (region A) represent the region where the solvents are strongly clustered by the hydrogen-bonding interactions. In region B represented with white color, however, the solute molecules destroy hydrogen bonding between the solvent molecules. Therefore, the solvent clustering effects are considered to be smaller in region B and solute molecules may diffuse in three-dimensional space. When the RPs are generated in region B, radicals do not penetrate region A because of the strong solvent–solvent interactions. As a result, region B can work as a three-dimensional cage in the one-dimensional pore. In the cage, the RPs diffuse and recombine, generating the MFEs. From the SLE analysis, the size of region B was estimated to be on the order of nanometers for both MCM-41. In region B, there are still the solvent clustering effects which give the viscosity higher than the bulk solution. When the cage parameters determined for MCM (2.7) and for MCM (1.7) are compared, one notices that the cage effects of MCM (2.7) are larger than those of MCM (1.7). Also η for MCM (2.7) is larger than η for MCM (1.7). These results suggested that the solvents in MCM (2.7) are

clustered stronger than those in MCM (1.7). So far, solvent clustering effects in mesoporous silicates have been reported with their diameters larger than 2 nm.^{14–16,21} In MCM (1.7), there may not be enough space for solvent clustering, resulting in the smaller cage effects observed in the present study.

Conclusion

The MFEs on photochemical reactions in 2-PrOH confined in mesoporous silicates of MCM-41 were studied under high magnetic fields up to 5 T. The observed MFEs were analyzed by using the SLE where the one-dimensional and three-dimensional diffusion models were employed. The SLE analysis revealed that the radical diffusions were not restricted in the one dimension, although the used MCM-41 has the one-dimensional nanopore with diameter of 2.7 or 1.7 nm. The effective viscosities of 2-propanol for the radical rotations in MCM-41 were higher than that for the bulk one. These results suggest clustering of solvent molecules in mesoporous silicates. In addition to that, our analysis indicates that the RPs are confined in the nanospace in MCM-41, although the nanopore is opened at both ends. The present results demonstrate that the MFEs study provides the information on molecular motions in nanospace.

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Supporting Information Available: Details of the sample characterization and the SLE analysis. This material is available free of charge via the Internet at <http://pubs.acs.org>.

References and Notes

- (1) Everett, D. H. *Pure Appl. Chem.* **1972**, *31*, 578–638.
- (2) Corma, A. *Chem. Rev.* **1997**, *97*, 2373–2419.
- (3) Taguchi, A.; Schüth, F. *Microporous Mesoporous Mater.* **2005**, *77*, 1–45.
- (4) Wight, A. P.; Davis, M. E. *Chem. Rev.* **2002**, *102*, 3589–3614.
- (5) De Vos, D. E.; Dams, M.; Sels, B. F.; Jacobs, P. A. *Chem. Rev.* **2002**, *102*, 3615–3640.
- (6) Soler-Illia, G. J. de A. A.; Sanchez, C.; Lebeau, B.; Patarin, J. *Chem. Rev.* **2002**, *102*, 4093–4138.
- (7) Kageyama, K.; Tamazawa, J.; Aida, T. *Science* **1999**, *285*, 2113–2115.
- (8) Ikegami, M.; Tajima, K.; Aida, T. *Angew. Chem., Int. Ed.* **2003**, *42*, 2154–2157.
- (9) Yiu, H. H. P.; Botting, C. H.; Botting, N. P.; Wright, P. A. *Phys. Chem. Chem. Phys.* **2001**, *3*, 2983–2985.
- (10) Yamaguchi, A.; Teramae, N. *Anal. Sci.* **2008**, *24*, 25–30.
- (11) Roy, I.; Ohulchanskyy, T. Y.; Bharali, D. J.; Pudavar, H. E.; Mistretta, R. A.; Kaur, N.; Prasad, P. N. *Proc. Natl. Acad. Sci. U.S.A.* **2005**, *102*, 279–284.
- (12) Ariga, K.; Vinu, A.; Hill, J. P.; Mori, T. *Coord. Chem. Rev.* **2007**, *251*, 2562–2591.
- (13) Jung, C.; Kirstein, J.; Platschek, B.; Bein, T.; Budde, M.; Framk, I.; Müllen, K.; Michaelis, J.; Bräuchle, C. *J. Am. Chem. Soc.* **2008**, *130*, 1638–1648.
- (14) (a) Yamaguchi, A.; Yoda, T.; Suzuki, S.; Morita, K.; Teramae, N. *Anal. Sci.* **2006**, *22*, 1501. (b) Kamijo, T.; Yamaguchi, A.; Suzuki, S.; Teramae, N.; Itoh, T.; Ikeda, T. *J. Phys. Chem. A* **2008**, *112*, 11535–11542.
- (15) (a) Takahara, S.; Nakano, M.; Kittaka, S.; Kuroda, Y.; Mori, T.; Hamano, H.; Yamaguchi, T. *J. Phys. Chem. B* **1999**, *103*, 5814–5819. (b) Takahara, S.; Kittaka, S.; Mori, T.; Kuroda, Y.; Takamuku, T.; Yamaguchi, T. *J. Phys. Chem. C* **2008**, *112*, 14385–14393.
- (16) (a) Okazaki, M.; Toriyama, K.; Sawaguchi, N.; Oda, T. *Appl. Magn. Reson.* **2003**, *23*, 435–444. (b) Okazaki, M.; Toriyama, K. *J. Phys. Chem. C* **2007**, *111*, 9122–9129. (c) Okazaki, M.; Iwamoto, S.; Sueishi, Y.;

- Toriyama, K. *J. Phys. Chem. C* **2008**, *112*, 768–775. (d) Okazaki, M.; Seelan, S.; Toriyama, K. *Appl. Magn. Reson.* **2009**, *35*, 363–378.
- (17) Steiner, U. E.; Ulrich, T. *Chem. Rev.* **1989**, *89*, 51–147.
- (18) Nagakura, S.; Hayashi, H.; Azumi, T. *Dynamic Spin Chemistry*; Kodansha-Wiley: Tokyo, 1998.
- (19) Hayashi, H. *Introduction to Dynamic Spin Chemistry*; World Scientific: Singapore, 2004.
- (20) (a) Wakasa, M. *J. Phys. Chem. B* **2007**, *111*, 9434–9436. (b) Hamasaki, A.; Yago, T.; Takamasu, T.; Kido, G.; Wakasa, M. *J. Phys. Chem. B* **2008**, *112*, 3375–3379. (c) Hamasaki, A.; Yago, T.; Wakasa, M. *J. Phys. Chem. B* **2008**, *112*, 14185–14192. (d) Wakasa, M.; Yago, T.; Hamasaki, A. *J. Phys. Chem. B* **2009**, *113*, 10559–10561.
- (21) (a) Okazaki, M.; Konishi, Y.; Toriyama, K. *Chem. Phys. Lett.* **2000**, *328*, 251–256. (b) Konishi, Y.; Okazaki, M.; Toriyama, K.; Kasai, T. *J. Phys. Chem. B* **2001**, *105*, 9101–9106. (c) Okazaki, M.; Toriyama, K.; Oda, K.; Kasai, T. *Phys. Chem. Chem. Phys.* **2002**, *4*, 1201–1205.
- (22) Yanagisawa, T.; Shimizu, T.; Kuroda, K.; Kato, C. *Bull. Chem. Soc. Jpn.* **1990**, *63*, 988–992.
- (23) Kresge, C. T.; Leonowicz, M. E.; Roth, W. J.; Vartuli, J. C.; Beck, J. S. *Nature* **1992**, *359*, 710–712.
- (24) Anandan, S.; Okazaki, M. *Microporous Mesoporous Mater.* **2005**, *87*, 77–92.
- (25) Namba, S.; Mochizuki, A.; Kito, M. *Chem. Lett.* **1998**, *27*, 569–570.
- (26) (a) Hamasaki, A.; Nishizawa, K.; Sakaguchi, Y.; Okada, T.; Kido, G.; Wakasa, M. *Chem. Lett.* **2005**, *34*, 1692–1693. (b) Hamasaki, A.; Sakaguchi, Y.; Nishizawa, K.; Kido, G.; Wakasa, M. *Mol. Phys.* **2006**, *104*, 1765–1771.
- (27) (a) Tanimoto, Y.; Takashima, M.; Itoh, M. *Chem. Phys. Lett.* **1983**, *100*, 442–444. (b) Tanimoto, Y.; Takashima, M.; Itoh, M. *J. Chem. Phys.* **1984**, *88*, 6053–6056.
- (28) Garner, A.; Wilkinson, F. *J. Chem. Soc., Faraday Trans. 2* **1976**, *72*, 1010–1020.
- (29) (a) Pedersen, J. B.; Freed, J. H. *J. Chem. Phys.* **1973**, *58*, 2746–2762. (b) Pedersen, J. B.; Freed, J. H. *J. Chem. Phys.* **1973**, *59*, 2869–2885.
- (30) Busmann, H.-G.; Staerk, H.; Weller, A. *J. Chem. Phys.* **1989**, *91*, 4098–4105.
- (31) Carrington, A.; McLachlan, A. D. *Introduction to Magnetic Resonance*; Harper, New York, 1967.
- (32) Steiner, U. E.; Wu, J. Q. *Chem. Phys.* **1992**, *162*, 53–67.
- (33) Maiti, N. C.; Krishna, M. M. G.; Britto, P. J.; Periasamy, N. *J. Phys. Chem. B* **1997**, *101*, 11051–11060.
- (34) Tanford, C. *J. Phys. Chem.* **1972**, *76*, 3020–3024.
- (35) Emert, J.; Behrens, C.; Goldenberg, M. *J. Am. Chem. Soc.* **1979**, *101*, 771–772.
- (36) Turro, N. J.; Aikawa, M.; Yekta, A. *J. Am. Chem. Soc.* **1979**, *101*, 772–774.
- (37) Turro, N. J.; Okubo, T. *J. Am. Chem. Soc.* **1981**, *103*, 7224–7228.
- (38) Turley, W. D.; Offen, H. W. *J. Phys. Chem.* **1985**, *89*, 2933–2937.
- (39) Shinitzky, M.; Dianox, A. C.; Gitter, C.; Webber, G. *Biochemistry* **1971**, *10*, 2106–2113.
- (40) Lukuzen, N. N.; Ivanov, K. L.; Morozov, V. A.; Kattnig, D. R.; Grampp, G. *Chem. Phys.* **2006**, *328*, 75–84.
- (41) Gjerdåker, L.; Aksnes, D. W.; Sørland, G. H.; Stöcker, M. *Microporous Mesoporous Mater.* **2001**, *42*, 89–96.
- (42) Hansen, E. W.; Courivaud, F.; Karlsson, A.; Kolboe, S.; Stöcker, M. *Microporous Mesoporous Mater.* **1998**, *22*, 309–320.

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